Mini Assignment 2

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Path 411: Applied Data Science in Molecular Medicine

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Quality of fastq file

HBR_Rep1_ERCC-Mix2_Build37-ErccTranscripts-chr22.read2
FastQC reports good quality for reads (key base-quality modules passed), FAIL in Per base sequence content (expected priming bias in RNA-seq) and passes in adapter content and length distribution. Trimming adapters is not absolutely required; light 3′ trimming optionally removed start-position composition bias. Can be retained for downstream analysis.

HBR_Rep2_ERCC-Mix2_Build37-ErccTranscripts-chr22.read1
Overall quality is good, with FAIL in Per base sequence content and passes for adapter content, length distribution, and overall quality modules. Composition bias at the beginning of reads is normal for RNA-seq and is not an exclusion; optional 3′ trimming can be performed for demanding pipelines. Can be considered for downstream analysis.

HBR_Rep2_ERCC-Mix2_Build37-ErccTranscripts-chr22.read2
There are FastQC modules in general PASS, e.g., adapter content, length distribution, base-quality scores. There is no corrective preprocessing proposed. Can be added to downstream analysis.

HBR_Rep3_ERCC-Mix2_Build37-ErccTranscripts-chr22.read1
Quality scores are fine with a FAIL in Per base sequence content and passing elsewhere (adapter content, length distribution, per-base quality). This is the normal pattern for priming bias and not contamination; trimming is optional. Can be added to downstream analysis.

HBR_Rep3_ERCC-Mix2_Build37-ErccTranscripts-chr22.read2
Nearly all modules PASS (per-base quality, basic statistics, per-tile quality, per-sequence quality, length distribution, N content, adapter content). No preprocessing is recommended. Can be carried forward in downstream analysis.

UHR_Rep1_ERCC-Mix1_Build37-ErccTranscripts-chr22.read1
Modules tested report passes on core quality(adapter/length checks - no critical failures identified). Trimming and filtering are not needed beyond standard pipeline defaults. Can be carried forward in downstream analysis.

UHR_Rep1_ERCC-Mix1_Build37-ErccTranscripts-chr22.read2
Core quality and adapter modules pass, with no major red flags in summary. No additional preprocessing required. Can be included in downstream analysis.

UHR_Rep2_ERCC-Mix1_Build37-ErccTranscripts-chr22.read1

Quality, adapter, and length modules pass, with no serious warnings observed. Preprocessing not required. Can be included in downstream analysis.

UHR_Rep2_ERCC-Mix1_Build37-ErccTranscripts-chr22.read2
FastQC indicates passes for adapter content and other primary checks, with no major quality issues. No trimming/filtering recommended. Can be included in downstream analysis.

UHR_Rep3_ERCC-Mix1_Build37-ErccTranscripts-chr22.read1
Most modules pass, with the exception of Sequence Duplication Levels, which is FAIL, as expected in RNA-seq due to highly expressed transcripts and is not necessarily indicative of technical failure. Proceed with reduced library complexity; no adapter trimming is recommended. Can be added to downstream analysis.

UHR_Rep3_ERCC-Mix1_Build37-ErccTranscripts-chr22.read2
Regarding paired reads, the majority of modules work with a FAIL in Sequence Duplication Levels consistent with biologically inspired duplication. No trimming of adapters is required; proceed and be aware of possible expression-driven duplication during quantification. Can be added to downstream analysis.

Across all samples, FastQC inspection shows overall per-base quality typically high and reasonable distributions of read lengths, with adapter content passing for all files; a few HBR reads show Per base sequence content failures as would be expected with RNA-seq priming bias, and UHR_Rep3 shows high duplication as would be expected for very highly expressed transcripts. No sample exclusion is warranted; trimming of 3′ ends of reads with base-composition bias on an optional basis can be done, and duplication flags in UHR_Rep3 can be viewed as biologically rather than technically. These results are congruent with the assignment requirements to assess inclusion and preprocessing choices from FastQC directly (e.g., trimming only when warranted).

Overall per-base quality and good read-length distributions are reported by FastQC in all samples, with adapter content passing for all the files; several HBR reads report Per base sequence content failures as would be expected of RNA-seq priming bias; and UHR_Rep3 has high duplication as would be expected for highly expressed transcripts. No sample exclusion is warranted; base composition bias reads may optionally be trimmed at the 3′ end, but UHR_Rep3 duplication flags need to be considered biologically, not technically. Such conclusions are in agreement with the assignment instruction of directly assessing inclusion and preprocessing options from FastQC (e.g., trimming only as warranted).

Ballgown directory results

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ENSG00000000003,TSPAN6,0,0,0,0,0,0,0
ENSG00000000005,TNMD,0,0,0,0,0,0,0
ENSG00000000419,DPM1,0,0,0,0,0,0,0
ENSG00000000457,SCYL3,0,0,0,0,0,0,0
ENSG00000000460,FIRRM,0,0,0,0,0,0,0
ENSG00000000938,FGR,0,0,0,0,0,0,0
ENSG00000000971,CFH,0,0,0,0,0,0,0
ENSG00000001036,FUCA2,0,0,0,0,0,0,0
ENSG00000001084,GCLC,0,0,0,0,0,0,0
ENSG00000001167,NFYA,0,0,0,0,0,0,0
ENSG00000001460,STPG1,0,0,0,0,0,0,0
ENSG00000001461,NIPAL3,0,0,0,0,0,0,0
ENSG00000001497,LAS1L,0,0,0,0,0,0,0
ENSG00000001561,ENPP4,0,0,0,0,0,0,0
ENSG00000001617,SEMA3F,0,0,0,0,0,0,0
ENSG00000001626,CFTR,0,0,0,0,0,0,0
ENSG00000001629,ANKIB1,0,0,0,0,0,0,0
ENSG00000001630,CYP51A1,0,0,0,0,0,0,0
ENSG00000001631,KRIT1,0,0,0,0,0,0,0
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In the six RNA-Seq samples, the numbers of total reads ranged from approximately 98,000 to 170,000, with HBR_Rep1–3 containing approximately 91,000–110,000 reads and UHR_Rep1–3 containing slightly more at 105,000–169,000 reads. There were 78,560 transcripts, although more than 99% of these had zero counts in all but a few samples and thus might have represented a very minor subgroup of genes which were actively expressed at detectable levels. After transcript excludance that was not detectably expressed, the average TPM (Transcripts Per Million) per sample was around 1,729, i.e., library-normalized gene abundance of expressed genes in a library.

Inside ballgown/, there are six central files in each sample subdirectory (e.g., HBR_Rep3_ERCC-Mix2) that are generated by StringTie and collectively represent transcript- and exon-level quantification. The t_data.ctab file holds transcript-level metrics such as IDs, lengths, coverage, and normalized counts (FPKM and TPM), while the e_data.ctab file holds exon-level coverage and coordinate data. The i_data.ctab and i2t.ctab files hold intron-level expression and intron-to-transcript mappings, and the e2t.ctab file creates exon-to-transcript mappings to support transcript reconstruction. The transcripts.gtf file holds structural annotations of assembled transcripts made up of genomic positions and gene affiliations.

Outside of sample-specific directories, the *_norm_cnts.tsv files contain normalized expression by sample, and the gene_count_matrix.csv and transcript_count_matrix.csv files contain all read counts aggregated by samples, which are raw inputs to Ballgown's differential expression analyses.